Sahaj Bhadja

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Summary

Software engineer with 2+ years building distributed API services, cloud workflows, and production automation. Experienced in Python, Java, TypeScript, REST APIs, microservices, Kubernetes, Docker, Jenkins, SQL, concurrency, performance tuning, testing, and operational reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected multi-tenant API and storage services for 500+ users, optimizing Azure SQL queries, indexes, and access paths to reduce control-plane dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant distributed workflows for 5,000+ monthly jobs using explicit states, retries, reconciliation checks, and failure isolation, tripling throughput while sustaining 99%+ production reliability under load.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Flask API services and React workflows supporting 150+ users and 10,000+ daily requests, integrating secure ingestion, normalized AWS RDS persistence, status tracking, and operational usage instrumentation.
- Engineered concurrent processing for 1,000+ files with automated validation, error handling, and fault-tolerant execution, reducing turnaround 60% while maintaining stable API performance during sustained peak workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible data and simulation pipelines processing 100,000+ records, applying structured testing, instrumentation, debugging, and performance analysis to improve long-horizon planning accuracy 18% across controlled evaluations.
- Collaborated on design reviews and technical documentation while tuning a performance-critical architecture, improving closed-loop stability 15% through reproducible benchmark comparisons and systematic investigation of complex failure scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Optimized distributed REST APIs using object-oriented patterns, filtering, pagination, indexing, and caching, improving response times 30% for bandwidth-constrained clients while increasing globally distributed ETL pipeline efficiency 68%.
- Automated nightly builds and 20+ unit and integration suites through Jenkins, PyTest, and Mockito across three operating systems, accelerating production defect diagnosis and improving distributed release reliability.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed eight modular microservices with REST APIs and WebSocket synchronization, supporting 2 million+ Azure SQL records across reliable customer-facing authentication, resource, customer, billing, permissions, and notification workflows.
- Automated Docker builds, tests, staged deployments, and production monitoring through Jenkins and Azure services, reducing release time 50% while improving overall long-term service maintainability and operational reliability.

Skills

Languages & APIs: Python, Java, C++, TypeScript, JavaScript, SQL, Bash, REST APIs, JSON, WebSockets, API design, object-oriented programming
Distributed Systems: microservices, concurrency, multithreading, asynchronous workflows, failure isolation, retries, caching, rate limiting, scalability, high availability
Cloud & Infrastructure: AWS, Azure, Kubernetes, Docker, Jenkins, CI/CD, GitHub Actions, S3, AWS Glue, Azure SQL, PostgreSQL, MySQL, MongoDB
Reliability & Delivery: automated testing, unit testing, integration testing, PyTest, Mockito, debugging, profiling, latency tuning, observability, incident analysis, code review, Agile, Scrum, production support

Projects

FinNexus | *TypeScript, WebSockets, MongoDB, Kubernetes, CI/CD*
- Built a scalable platform supporting 50,000+ users and one million transactions, integrating secure APIs, real-time synchronization, MongoDB persistence, Kubernetes deployment, and automated CI/CD release workflows.

Chat2SQL | *Java, TypeScript, PostgreSQL, MySQL, MongoDB, Docker*
- Built a cross-platform workspace supporting 10+ relational and NoSQL storage systems, with schema-aware query generation, structured validation, optimized execution, and Docker-backed deployment across Java and TypeScript services.

Big-Brain | *Python, PostgreSQL, GCP, REST APIs, FAISS*
- Engineered a cloud data ingestion and indexing pipeline processing 50,000+ records with incremental updates, metadata persistence, and performance tuning, reducing retrieval latency 40% through optimized execution paths.